

initial

Decorative initial capitals

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Table 1: Engines and formats with which initial was tested; note that a scalable font is necessary

Abstract

The initial package provides decorative initial letters for L^AT_EX, derived from `lettrine.sty` and `novel.cls`, but standalone. Core features include a three-region paragraph shaping model with independent control over first-line, initial-region, and post-initial formatting; a preset system supporting named configurations with inheritance; automatic vertical clearance to prevent collision with preceding text; native color and contour effects; diacritic handling. The package detects image files automatically, supports PDF accessibility tagging through the same functionality in `lettrine`, and provides baseline grid alignment for strict typographic applications. Parameter vocabulary deliberately follows `lettrine`, where functionality overlaps. This documentation includes both technical reference and an extended historical survey of initial letter design from Insular manuscripts through digital typography.

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1 Introduction

```
% main package
\usepackage{initial}

% custom preset behavior
\newinitial{presetname}{options}

% text beginning with decorative initial
\initial{T}he \cmd{initial} command creates a drop cap spanning the
    specified number of lines. The first argument (optional) controls
    appearance through a number of options, most of which are key-value
    pairs; the second is the letter to be the initial; the third
    argument (optional) is text after the initial to be stylized;
    everything after flows around it. To define reusable presets, set
    with \cmd{newinitial}\{\opt{name}\}\{\opt{options}\}. Image files (.
    pdf, .png, .jpg) are detected automatically. The package handles
    paragraph shaping, baseline alignment, and collision prevention
    without manual intervention.
```

The `\initial` command creates a dropped capital spanning the specified number of lines. The first argument (optional) controls appearance through a number of options, most of which are key-value pairs; the second is the letter to be the initial; the third argument (optional) is text after the initial to be stylized; everything after flows around it. To define reusable presets, set with `\newinitial{name}{options}`. Image files (.pdf, .png, .jpg) are detected automatically. The package handles paragraph shaping, baseline alignment, and collision prevention without manual intervention.

```
\initial[<options>]{<letter>}[<text>]
```

The file `initial.sty` provides the `\initial` command; no options are passed to the package, and all formatting is controlled by the command. The command accepts optional arguments to control the size and layout of the required letter argument and subsequent paragraph behavior, with optional text to be formatted appropriately.

1.1 Reinventing the wheel

It is good practice to avoid reinventing the wheel and so expend effort to reproduce existing solutions; stronger yet, good practice cautions against reinventing the wheel as a square, where the reproduction functions worse than what it purports to replace. `initial.sty` does both.

Daniel Flipo's `lettrine.sty` has provided L^AT_EX users with reliable drop cap functionality since 1999; indeed, over twenty-five years of refinement have produced a package of significant completeness. For production work, `lettrine` remains the correct choice; it is time-tested, widely documented, and integrated into

countless existing workflows. Nothing in `initial.sty` improves upon `lettrine`'s core capabilities. `initial.sty` is a square wheel.

1.2 Plagiarism and pedagogy

From T.S. Eliot:

Immature poets imitate; mature poets steal.

The distinction is in transformation: imitation reproduces surface features; theft assimilates and reconstitutes. Contrast with W.H. Davenport Adams, who earlier noted:

Great poets imitate and improve, whereas small ones steal and spoil.

Both acknowledge that originality emerges through engagement with predecessors rather than isolation from them.

`initial.sty` steals with abandon. The parameter names `findent`, `nindent`, and `slope`, among others, come directly from `lettrine`. The `ante` mechanism, the image detection logic, the fundamental approach of `\parshape` construction followed by `\llap` placement all derive from studying Flipo's implementation and Robert Allgeyer's `novel.cls` adaptations. Where `initial.sty` diverges (the presets, the preset factory, style toggles, three-region paragraph model, vertical clearance automation), these represent not improvements (!), but offer alternative approaches explored for their own sake, which we can only hope may be of interest, or even some use.

The motivation for this package was primarily pedagogical; $\text{\LaTeX}$'s paragraph-shaping primitives, box manipulation commands, and font metric calculations are a domain where reading documentation provides incomplete understanding. The machinery becomes clear only through development: discovering why and how `\smash` must wrap the initial to prevent line spacing disruption; why and how height computation requires iterative adjustment against actual rendered output; why and how `\llap` positions correctly where `\hbox` to 0pt would not. Indeed, these lessons transfer to other typographic problems, regardless of whether `initial.sty` (or earlier, messier versions of it) itself sees use.

Drop caps run against multiple $\text{\TeX}$ subsystems: paragraph breaking; box construction; font selection; dimension arithmetic; macro expansion. A working implementation requires each subsystem to cooperate; failure to do so is equally interesting, which might otherwise be the case that successful operation obscures. Therefore, learning about such systems by way of interesting typography seems a natural motivation; this is as much an exercise in the Feynman technique as it is constructing an in-house drop cap functionality for my own work as it is contributing to the $\text{\LaTeX}$ community as it is simply a fun thing to do; and let fun be not the least of things.

1.3 Acknowledgments

Primary acknowledgment belongs to Daniel Flipo, whose `lettrine.sty` established the standard this package studies and imitates. The parameter vocabulary, the `\parshape` approach, the slope mechanism for diagonal letters, the grid accountability; `lettrine` provided the template. Flipo’s continued maintenance through version 2.83 (December 2025) demonstrates the sustained commitment that production packages require.

Robert Allgeyer’s `novel.cls` contributed the specific adaptations for fiction publishing that motivated several features: strict baseline grid awareness; emphasis on PDF/X compliance; the philosophical stance that drop caps serve documents rather than demanding documents serve them.

The CTAN maintainers and T_EX community preserve and distribute both packages, along with predecessors (`dropcaps.sty`; `drop.sty`; `dropping.sty`) whose evolution `lettrine` synthesized. Mats Dahlgren’s `dropping.sty` (1997) provided early exploration of the problem space, demonstrating approaches `lettrine` later refined or resolved.

Whatever errors, inefficiencies, and misjudgments `initial.sty` introduces beyond its sources are our own.

2 Package details

The package requires L^AT_EX2e, and uses standard packages.

- | | |
|-------------------------|------------------------------|
| • <code>xparse</code> | document command definitions |
| • <code>keyval</code> | key-value option parsing |
| • <code>xstring</code> | string manipulation |
| • <code>ifthen</code> | conditional logic |
| • <code>graphicx</code> | image inclusion and scaling |
| • <code>calc</code> | dimension arithmetic |
| • <code>etoolbox</code> | macro building utilities |
| • <code>xcolor</code> | color support |
| • <code>contour</code> | contour/outline effects |

The package accepts no loading options; all configuration happens either through the option system when calling `\initial`, or by defining styles or setting global defaults in the preamble.

`\initial[options]{character or image}[optional post-initial text]`

The `\initial` command must appear at the start of a paragraph; everything following the closing brace becomes the paragraph text, flowing around the initial according to the paragraph shaping the package computes. The paragraph terminates (normally) in a blank line, or explicit `\par` ends it, at which point normal formatting resumes.

```
\initial{L}orem ipsum dolor sit amet, consectetur adipiscing elit.
    Maecenas pellentesque, orci vitae blandit pharetra, orci ligula
    finibus diam, sed tempus erat tortor id magna. Pellentesque habitant
    morbi tristique senectus et netus et malesuada fames ac turpis
    egestas.

This is a new paragraph with no initial.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

This is a new paragraph with no initial.

The package uses `\llap` (left overlap) to place the initial, in which the character does not consume horizontal space in the normal text flow; the package computes appropriate `\parshape` values that indent the text rightward, creating room for the initial to occupy. The initial itself, then, floats in the created space such that it is visually anchored, but typographically separate, from the paragraph's geometry.

2.1 A note on images

When the argument contains a recognized image extension in `.pdf`, `.png`, `.jpg`, or `.jpeg` (matched case-insensitively), the package switches from typesetting a glyph to including an image via `\includegraphics`. The image scales to match the computed initial height while preserving its aspect ratio.

```
\initial[actualtext=A]{deca.png}ve atque vale. Lorem ipsum dolor sit amet
    , consectetur adipiscing elit. Maecenas pellentesque, orci vitae
    blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
    id magna. Pellentesque habitant morbi tristique senectus et netus et
    malesuada fames ac turpis egestas.
```

ve atque vale. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Daniel Flipo solved the problem of integrating accessibility when using an image as the argument of the initial; we implement that here with `actualtext` (which defaults to whatever is the argument of the command if that argument is already text); indeed, when typesetting a letter, the character itself provides semantic content for screen readers and PDF text extraction. An image carries

no such inherent meaning, so the `actualtext` option addresses this by specifying the replacement text.

If the image is used without the `actualtext` stated, the package issues a warning:

Package initial Warning: no actualtext provided for image initial.
 For accessibility, use `actualtext=<letter>` option.
 Example: `\initial[actualtext=W]{ornamental-W.pdf}`

2.2 Dimensional parameters

See Figure 1 for the physical features being manipulated in the package.

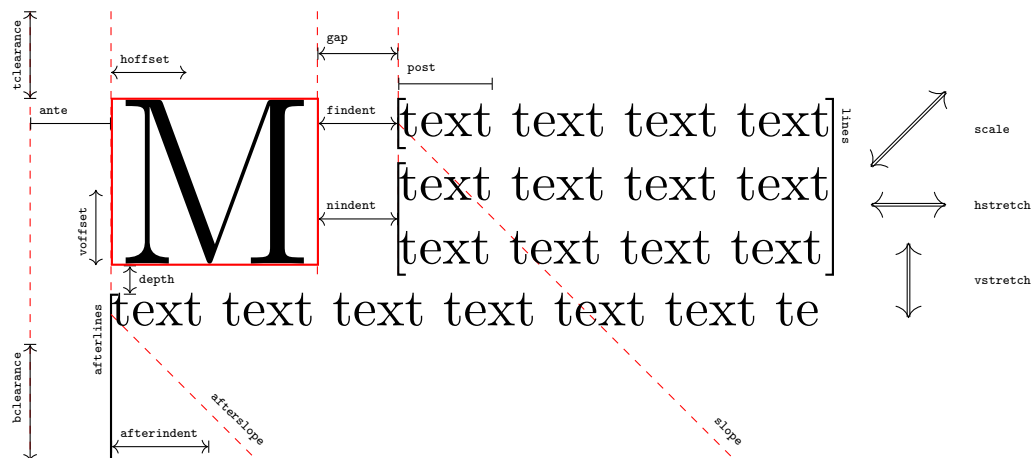


Figure 1: Anatomy of the initial package options; note that, for clarity, options `scale`, `hstretch`, and `vstretch` are visually displaced to the side, which are options that directly operate on the initial object in the red bounded box.

The dimensional parameters control where the initial sits and how text flows around it. Drop cap positioning involves two independent concerns: where the initial itself appears; how the paragraph indents to accommodate it. The offset parameters (`hoffset`, `voffset`) control the former; the indentation parameters (`findent`, `nindent`, `gap`), the latter.

Option	Type	Default	Description	Global override
<code>lines</code>	integer	3	lines initial spans vertically	<code>\initialDefaultLines</code>
<code>depth</code>	integer	0	additional lines below initial region	<code>\initialDefaultDepth</code>
<code>hoffset</code>	length	0pt	horizontal shift	<code>\initialDefaultHoffset</code>
<code>voffset</code>	length	0pt	vertical shift	<code>\initialDefaultVoffset</code>
<code>findent</code>	length	0pt	first line gap	<code>\initialDefaultFindent</code>
<code>nindent</code>	length	0pt	subsequent lines gap	<code>\initialDefaultNindent</code>
<code>gap</code>	length	0pt	additive shift to both	—
<code>slope</code>	length	0pt	per-line indent increment	<code>\initialDefaultSlope</code>
<code>scale</code>	decimal	1	overall size multiplier	<code>\initialDefaultScale</code>
<code>hstretch</code>	float	1	horizontal stretch factor	<code>\initialDefaultHstretch</code>
<code>vstretch</code>	float	1	vertical stretch factor	<code>\initialDefaultVstretch</code>

Lines and depth The `lines` parameter specifies how many text lines the initial spans vertically (default 3). The package computes height by multiplying the baseline skip by one less than the line count, then adding the cap height of the reference character. The initial’s top aligns with the first line’s cap height; its bottom rests on the N th baseline.

```
\initial[lines=4]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

When `lines=1`, the package bypasses the full `\parshape` machinery: the initial renders inline at the paragraph’s start, followed by any post text and the paragraph body. No paragraph shaping occurs; the initial simply occupies its natural width plus `findent` spacing.

```
\initial[lines=1]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The single-line path ignores `nindent`, `slope`, `depth`, and the after-region parameters, as these govern multi-line shaping that does not apply. The `ante`, `font`, `style` toggles, and `color` options function normally.

The `depth` parameter extends paragraph shaping beyond the initial's visual extent. Setting `depth=1` adds one more line of indentation below where the initial ends; this accommodates letters with descenders such as <Q>, drop below the baseline, and their tails need clearance from the text flowing beside them. Any decorative flourishes extending below the letter body similarly benefit from depth adjustment.

```
\initial[depth=1]{Q}uis custodiet ipsos custodes. Lorem ipsum dolor sit
    amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
    blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
    id magna. Pellentesque habitant morbi tristique senectus et netus et
    malesuada fames ac turpis egestas.
```

Q uis custodiet ipsos custodes. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Horizontal and vertical offsets The `hoffset` parameter shifts the initial horizontally from its default position: positive values push it rightward into the text block; negative values leftward into the margin.

```
\initial[hoffset=-2.6em]{L}orem ipsum dolor sit amet, consectetur
    adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
    orci ligula finibus diam, sed tempus erat tortor id magna.
    Pellentesque habitant morbi tristique senectus et netus et malesuada
    fames ac turpis egestas.
```

L orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `voffset` parameter shifts the initial vertically: positive values raise it; negative values lower it. Vertical offsets apply after the standard positioning calculation, technically making them adjustments as opposed to absolute coordinates.

```
\initial[voffset=1.5\baselineskip]{L}orem ipsum dolor sit amet,
    consectetur adipiscing elit. Maecenas pellentesque, orci vitae
    blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
    id magna. Pellentesque habitant morbi tristique senectus et netus et
    malesuada fames ac turpis egestas.
```

L orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

First- and subsequent-line indentation The `findent` parameter controls the horizontal gap between the initial and the first line of text; at zero (the default), text begins immediately adjacent to the initial's right edge, which certainly may be too tight for comfortable reading, depending on the letter shape and font. Increasing `findent` opens breathing room:

```
\initial[findent=1.5em]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `nindent` parameter serves the same purpose, but for lines 2 through `lines + depth`. Together, `findent` and `nindent` define the rectangular indentation region around the initial. Setting them equal produces uniform spacing; setting `findent` larger than `nindent` creates a first line that sits further from the initial than subsequent lines.

The `gap` parameter provides for adjusting both indents simultaneously, but additively in total, such that there is a base gap that preserves any differential between first and subsequent lines:

```
\initial[findent=1.5em, gap=1.5em]{L}orem ipsum dolor sit amet,
consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Slope A strict rectangular indentation works poorly for diagonal letters: an `<A>` widens toward the bottom, so wrapping text in a rectangle wastes the space along its right diagonal. A `<V>` narrows toward the bottom; rectangular indentation leaves excessive space below. The `slope` parameter addresses diagonal letters by adding incremental indentation to each line.

```
\initial[lines=4, slope=0.5em, gap=-0.75em]{A}lea iacta est. Lorem ipsum
dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque,
orci vitae blandit pharetra, orci ligula finibus diam, sed tempus
erat tortor id magna. Pellentesque habitant morbi tristique senectus
et netus et malesuada fames ac turpis egestas.
```

Alea iacta est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

For letters that narrow toward the bottom rather than widening, negative slope pulls subsequent lines leftward:

```
\initial[lines=4, slope=-0.5em,gap=0.25em]{V}ale decem. Lorem ipsum dolor
sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci
vitae blandit pharetra, orci ligula finibus diam, sed tempus erat
tortor id magna. Pellentesque habitant morbi tristique senectus et
netus et malesuada fames ac turpis egestas.
```

Vale decem. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Proper slope values depend on the specific font's letter geometry. There is no universal setting; each typeface requires calibration.

Scale and stretch The `scale` parameter multiplies the computed height uniformly. A value of 1.2 produces an initial 20% larger than the lines parameter alone would yield; 0.9 produces one 10% smaller.

```
\initial[scale=1.5]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

For non-uniform scaling, `hstretch` and `vstretch` apply independent horizontal and vertical factors through `\scalebox`, sometimes resulting in silly warped letters. These parameters combine multiplicatively with `scale`, enabling compound transformations when needed.

```
\initial[scale=1.5,hstretch=1.5,hstretch=2.5]{L}orem ipsum dolor sit amet
, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci

ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

```
\initial[scale=1.5,vstretch=1.5,hstretch=0.5]{L}orem ipsum dolor sit amet
, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

2.3 Three-region paragraph model

Technically, the package controls for paragraph shaping in addition to the characteristic appearance of the initial; it does this through three distinct regions, each with independent control. The idea is to enable formatting patterns that would otherwise require manual `\parshape` construction.

The first region consists solely of the first line. Its indentation equals the initial's width plus `hoffset` plus `findent`, creating the characteristic gap beside the initial's top. When ante text exists inline, as opposed to hanging in the margin, its width adds to this calculation.

The second region spans lines 2 through `lines + depth`; these lines use `nindent` as their base indentation, with `slope` accumulating incrementally. This region shapes text around the body of the initial.

The third region activates when `afterlines` exceeds zero; the package then generates 100 additional shaped lines (effectively infinite for any practical paragraph) using `afterindent` as the base and `afterslope` as the per-line increment; this enables hanging indentation patterns where text following the initial region maintains consistent indentation:

```
\initial[afterlines=1, afterindent=4em]{L}orem ipsum dolor sit amet,
consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas. Vestibulum ante ipsum primis in
faucibus orci luctus et ultrices posuere cubilia curae; Vestibulum
scelerisque orci ac mollis mattis. Donec cursus suscipit lectus,
eget molestie erat placerat ut. Aliquam lobortis justo massa.
Integer vulputate est sem, quis auctor neque efficitur vehicula.
Nunc et dignissim ex.
```


Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Vestibulum scelerisque orci ac mollis mattis. Donec cursus suscipit lectus, eget molestie erat placerat ut. Aliquam lobortis justo massa. Integer vulputate est sem, quis auctor neque efficitur vehicula. Nunc et dignissim ex.

The `afterslope` parameter works like `slope`, but within this third region, enabling gradual return toward the margin or other transitional effects:

```
\initial[afterlines=1, afterindent=3em, afterslope=-0.5em]{L}orem ipsum
dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque,
orci vitae blandit pharetra, orci ligula finibus diam, sed tempus
erat tortor id magna. Pellentesque habitant morbi tristique senectus
et netus et malesuada fames ac turpis egestas. Vestibulum ante
ipsum primis in faucibus orci luctus et ultrices posuere cubilia
curae; Vestibulum scelerisque orci ac mollis mattis. Donec cursus
suscipit lectus, eget molestie erat placerat ut. Aliquam lobortis
justo massa. Integer vulputate est sem, quis auctor neque efficitur
vehicula. Nunc et dignissim ex.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Vestibulum scelerisque orci ac mollis mattis. Donec cursus suscipit lectus, eget molestie erat placerat ut. Aliquam lobortis justo massa. Integer vulputate est sem, quis auctor neque efficitur vehicula. Nunc et dignissim ex.

Option	Type	Default	Description	Global override
<code>afterlines</code>	integer	0	flag: enables post-initial control	<code>\initialDefaultAfterlines</code>
<code>afterindent</code>	length	0pt	post-initial base indent	<code>\initialDefaultAfterindent</code>
<code>afterslope</code>	length	0pt	post-initial region increment	<code>\initialDefaultAfterslope</code>

2.4 Typography controls

The package offers a number of controls over the behavior and appearance of certain typographic features. To begin, make sure that a scalable font is being used, otherwise control over its typography will be lost.

Option	Type	Default	Description	Global Override
font	command	—	font selection	—
bold	boolean	false	apply <code>\bfseries</code>	—
italic	boolean	false	apply <code>\itshape</code>	—
smallcaps	boolean	false	apply <code>\scshape</code>	—
slanted	boolean	false	apply <code>\slshape</code>	—
color	color name	(inherit)	initial color	<code>\initialDefaultColor</code>
contour	boolean	false	enable contour effect	—
contourcolor	color name	white	contour stroke color	<code>\initialDefaultContourColor</code>
contoursize	length	0.3pt	contour stroke width	<code>\initialDefaultContourSize</code>

Font selection The `font` parameter accepts any font-switching command. For documents using `fontspec` with `LuaLaTeX` or `XeLaTeX`, this enables complete typographic control over the initial’s appearance:

```
\initial[font=\sffamily]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

```
\initial[font=\unclfamily]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

```
\initial[font={\fontspec{EB Garamond}[Numbers=OldStyle] }]{L}orem ipsum
dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque,
orci vitae blandit pharetra, orci ligula finibus diam, sed tempus
erat tortor id magna. Pellentesque habitant morbi tristique senectus
et netus et malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Font styles Four boolean parameters control standard font styles and combine freely, provided the font supports that combination: the `bold` toggle applies `\bfseries`; `italic` applies `\itshape`; `smallcaps` applies `\scshape`; `slanted` applies `\slshape`.

The toggles build a style macro internally, applying after whatever the font parameter establishes. Writing `font=\sffamily`, `bold` produces a bold sans-serif initial: the font switch happens first, then the bold weight applies.

```
\initial[bold, italic]{L}orem ipsum dolor sit amet, consectetur  
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,  
orci ligula finibus diam, sed tempus erat tortor id magna.  
Pellentesque habitant morbi tristique senectus et netus et malesuada  
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

```
\initial[font=\sffamily, bold]{L}orem ipsum dolor sit amet, consectetur  
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,  
orci ligula finibus diam, sed tempus erat tortor id magna.  
Pellentesque habitant morbi tristique senectus et netus et malesuada  
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Color The `color` parameter accepts any color name that `xcolor` recognizes; this includes the predefined names, colors defined via `\definecolor`, and inline color model specifications:

```
\initial[color=red]{L}orem ipsum dolor sit amet, consectetur adipiscing  
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci  
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque  
habitant morbi tristique senectus et netus et malesuada fames ac  
turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

An empty color value is the default, and inherits whatever color the surrounding text of the document uses.

Contour effects The package loads the nifty `contour` package. The `contour` boolean enables outline rendering around the initial character. The `contour`

package produces this effect by overprinting the character multiple times with slight offsets in the contour color, then printing the character itself on top. The result resembles a halo or outline surrounding the letterform. The `contourcolor` parameter specifies the outline color (default white), and `contoursize` controls thickness (default 0.3pt). Larger contour sizes create more pronounced outlines, but may overwhelm otherwise fine letter details.

```
\initial[contour, contourcolor=red,contoursize=1pt]{L}orem ipsum dolor
sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci
vitae blandit pharetra, orci ligula finibus diam, sed tempus erat
tortor id magna. Pellentesque habitant morbi tristique senectus et
netus et malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Note: some fonts render contours poorly, particularly at the intersection of strokes where the overprinting creates visible artifacts; proceed with caution.

Behavior in lists The package accounts for some indented environments by incorporating `\rightmargin` into its `\parshape` calculations. Within *quote*, the initial and shaped text respect the environment’s margins rather than extending to the page edge.

```
\begin{quote}
\initial{L}orem ipsum dolor sit amet, consectetur adipiscing elit.
Maecenas pellentesque, orci vitae blandit pharetra, orci ligula
finibus diam, sed tempus erat tortor id magna. Pellentesque habitant
morbi tristique senectus et netus et malesuada fames ac turpis
egestas.
\end{quote}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

This behavior does not hold in lists, however, like *itemize* and *enumerate*; proceed with caution.

```
\begin{itemize}
\item \initial{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

```

\item \initial{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
\end{itemize}

```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

2.5 Ante and post text

Opening quotation marks, section symbols, or other punctuation or notation sometimes precede the initial letter; the `ante` and `ante*` parameters handle this placement.

Traditional typography tends to render the first word, line, or syllable following the initial in small capitals or other distinguished styling (this is a requirement of the `lettrine` package, for comparison). This creates a visual connection between the prominent initial and the body text, easing the reader's transition. The `post` parameter and trailing optional argument handle this styling.

Option	Type	Description
<code>ante=<i>text</i></code>	string	text before initial; 70% of initial height, raised, in margin
<code>ante*=<i>text</i></code>	string	text before initial; body size, at baseline, hangs in margin
<code>post=<i>style</i></code>	L ^A T _E X font style	processes style for optional post argument after initial

Ante text The `ante` parameter places text in the left margin, scaled to approximately match the initial's visual weight. The scaling uses 70% of the initial's computed height as reference, producing ante text smaller than the initial, but substantially larger than body text. A 0.2em gap separates the ante from the initial:

```

\initial[ante={`}]Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.

```

“**L**orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The **ante*** parameter (note the asterisk!) places text in the margin at body text size, without scaling; this preserves the quotation mark’s normal appearance while still positioning it outside the text block:

```
\initial[ante*={``}]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

“**L**orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Both variants hang the ante text into the margin; it does not affect `\parshape` calculations. The indentation computes from the initial alone, and so marginal matter exists outside the text block’s geometry.

Note the fixed 0.2em gap between the ante and the initial: this spacing is not configurable; it represents a typographic judgment about the visual separation between marginal punctuation and the initial letterform. For cases requiring different spacing, the ante text itself can include explicit kerning:

```
\initial[ante={``\kern1.5em}]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

“

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Post text The trailing optional argument provides the text to be styled. By default, this text renders in small capitals:

```
\initial{L}[orem ipsum] dolor sit amet, consectetur adipiscing elit.
Maecenas pellentesque, orci vitae blandit pharetra, orci ligula
finibus diam, sed tempus erat tortor id magna. Pellentesque habitant
morbi tristique senectus et netus et malesuada fames ac turpis
egestas.
```

LOREM IPSUM dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus

erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `post` parameter controls the styling applied to this text. Available styles mirror standard L^AT_EX font commands: `textsc` (small capitals); `textbf` (bold); `textit` (italic); `textsl` (slanted); `texttt` (monospace). Use `none` or `normal` for unstyled text:

```
\initial[post=textbf]{L}[orem ipsum] dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Multiple styles combine with comma separation; braces are required when multiple values appear within the options. Note that composition of stylings is only successful if the font supports it.

```
\initial[post={textit,textbf}]{L}[orem ipsum] dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

*L*orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The default styling can be changed globally via `\initialDefaultPost`:

```
% bold small caps default
\renewcommand*{\initialDefaultPost}{textsc,textbf}

% no styling by default
\renewcommand*{\initialDefaultPost}{none}
```

When no trailing argument is provided, no post text appears, and the paragraph continues normally.

2.6 Vertical clearance

A large initial extending above the first line's cap height could well collide with preceding text unless something prevents it from doing so; the package automatically calculates whether collision would occur, and adds vertical space when needed. The formula accounts for the initial's height, any vertical offset,

and includes a 0.3 baseline cushion to prevent uncomfortably tight spacing even when collision would not technically occur.

This automation handles the common case transparently. A scaled initial following a normal paragraph receives appropriate space without explicit intervention:

Regular paragraph text ending here.

```
\initial[lines=4,scale=1.25]{L}[orem ipsum] dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Regular paragraph text ending here.

LOREM IPSUM dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Option	Type	Default	Description	Global override
<code>tclearance</code>	length	Opt	fixed space above	<code>\initialDefaultTclearance</code>
<code>bclearance</code>	length	Opt	fixed space below	<code>\initialDefaultBclearance</code>
<code>textlines</code>	integer	0	paragraph line count	—

Explicit top and bottom clearance The `tclearance` parameter adds explicit space above the initial paragraph, supplementing automatic collision prevention. This suits chapter openings or section breaks where visual separation should exceed the minimum needed to prevent collision:

```
\initial[lines=4,tclearance=0.5\baselineskip]{L}[orem ipsum] dolor sit
amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

LOREM IPSUM dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `bclearance` parameter adds space after the paragraph ends. This proves useful when the initial's influence region extends beyond the paragraph's text lines.

```
\initial[lines=4,bclearance=0.5\baselineskip]{L}[orem ipsum] dolor sit
amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
```



```
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

LOREM IPSUM dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Calculated spacing with textlines When the initial spans more lines than the paragraph contains, subsequent text would collide with the initial's lower portion without corrective spacing. The `textlines` parameter declares how many lines the paragraph contains, enabling the package to calculate exactly how much bottom spacing is needed.

```
\initial[lines=4, textlines=1]{A}ve atque vale.
```

Next paragraph starts after calculated space.

Ave atque vale.

Next paragraph starts after calculated space.

The formula computes

$$\text{space} = \max(0, (\text{lines} + \text{depth} - \text{textlines})) \times \text{baselineskip}$$

```
\initial[lines=5, textlines=4]{A}ve atque vale. Maecenas pellentesque,
orci vitae blandit pharetra, orci ligula finibus diam, sed tempus
erat tortor id magna. Pellentesque habitant morbi tristique senectus
et netus et malesupta fames ac turpis egestas.
```

Ave atque vale. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesupta fames ac turpis egestas.

The `textlines` and `bclearance` parameters combine additively.

```
\initial[lines=4, textlines=1, bclearance=0.5\baselineskip]{A}ve atque
vale.
```

Next paragraph starts after calculated space.

Ave atque vale.

Next paragraph starts after calculated space.

Grid In grid typesetting, all baselines across a spread align to a common vertical grid, taken to be at `\baselineskip` intervals; this alignment creates agreement between facing pages, and is particularly important in multi-column layouts. The vertical space added above an initial paragraph can disrupt this grid if it does not equal an integer multiple of `\baselineskip`.

By default, `initial` adds exactly the space needed to prevent collision with the preceding paragraph. This calculated value is based on initial height, vertical offset, and clearance settings, and rarely (if only accidentally) lands on a grid line:

```
\initial[lines=4]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `grid` parameter rounds this vertical space up to the nearest integer multiple of `\baselineskip`, restoring grid alignment for all subsequent text:

```
\initial[lines=4, grid]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Note that additional whitespace is introduced above the initial paragraph; if the calculated space is $1.3\text{\baselineskip}$, grid mode rounds to $2\text{\baselineskip}$, which should be visually acceptable in most contexts, but noticeable with large initials or generous clearances.

For documents that require strict grid alignment throughout, enable grid mode globally:

```
\initialDefaultGridtrue
```

This setting applies to all subsequent `\initial` calls, unless overridden with `grid=false` on individual invocations.

Note: grid rounding applies only to the vertical space above the initial paragraph; the `tclearance` parameter adds to this space before rounding occurs, so explicit clearance values also benefit from grid alignment. The `bclearance` and `textlines` parameters, which control space below the paragraph, are not

subject to grid rounding; if grid-aligned bottom spacing is required, specify these values as explicit multiples of `\baselineskip`.

2.7 Diacritic handling

By default, the initial height calculation measures the actual character (including diacritics like `<Ä>`, `<Ö>`, `<Ñ>`) to prevent clipping. Letters with diacritics appear slightly smaller than their counterparts when given the same `lines` value.

The `basecap` option provides an alternative typographic treatment.

Mode	Height basis	Diacritic handling
<code>basecap=false</code>	full character <code><Ä></code>	contained within line span
<code>basecap=true</code>	base cap height <code><X></code>	overflows above line span

When `basecap=true`, the diacritic extends above the initial region as decorative ornamentation. The collision prevention algorithm uses the *actual rendered height* (including the diacritic), so the standard 0.3-`baselineskip` cushion applies relative to the diacritic's top edge, such that no additional padding is needed. In this way, an `<Ö>` with `basecap` will have the same base letter size as an `<O>` without.

<code>\initial{Ö}sterreich liegt in Mitteleuropa. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.</code>
<code>\initial[basecap]{Ö}sterreich liegt in Mitteleuropa. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.</code>

Österreich liegt in Mitteleuropa. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Österreich liegt in Mitteleuropa. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Automatic collision prevention still applies, but now relative to the diacritic's top edge rather than the nominal region. Preceding text remains clear, even when diacritics overflow upward.

2.8 Preset system

Consistent initial formatting across a document requires repeating the same options at every `\initial` call, which, after working on the package for some

time, has been found to be a tedious and error-prone endeavor. The `preset` system addresses this by allowing named option bundles that can be invoked with a single reference.

User presets The `\newinitial` command defines a preset by associating a name with a set of options, and subsequent `\initial` calls invoke this preset by name.

```
\newinitial{chapter}{%
  lines=4,
  gap=0.2em,
  color=red,
  bold,
  tclearance=0.5\baselineskip
}

\initial[preset=chapter]{L}orem ipsum dolor sit amet, consectetur
  adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
  orci ligula finibus diam, sed tempus erat tortor id magna.
  Pellentesque habitant morbi tristique senectus et netus et malesuada
  fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

All options from the preset apply as though written directly. A document might well define several presets for different contexts, like chapter openings, section beginnings, epigraph paragraphs, and select among them throughout as needed.

When a preset is active, user-specified options modify the preset’s baseline rather than replacing it. The package implements this via a two-pass option parsing:

1. first pass captures all user options;
2. apply any specified preset;
3. second pass: re-apply user options atop preset values.

```
\newinitial{base}{lines=4, color=blue, bold}

\initial[preset=base]{L}orem ipsum dolor sit amet, consectetur adipiscing
  elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
  ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
  habitant morbi tristique senectus et netus et malesuada fames ac
  turpis egestas.
```

```
\initial[preset=base, color=red]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

```
\initial[preset=base, lines=5, italic]{L}orem ipsum dolor sit amet,
consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

This enables variations on a theme without duplicating the common settings.

Furthermore, named presets can reference other presets, enabling inheritance hierarchies:

```
\newinitial{base}{lines=3, gap=0.2em, bold}
\newinitial{fancy}{preset=base, color=purple, italic}
\newinitial{fancier}{preset=fancy, contour, contourcolor=green}

\initial[preset=fancier]{L}orem ipsum dolor sit amet, consectetur
adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
orci ligula finibus diam, sed tempus erat tortor id magna.
Pellentesque habitant morbi tristique senectus et netus et malesuada
fames ac turpis egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Here *fancier* inherits from *fancy*, which inherits from *base*; the final effect accumulates all three levels. The package tracks recursion depth to prevent infinite loops from circular references: after 5 levels of preset expansion, it issues an error, rather than continuing indefinitely:

```

% this would error: circular reference
\newinitial{loop1}{preset=loop2, color=red}
\newinitial{loop2}{preset=loop1, color=blue}

\initial[preset=loop1]{L}orem ipsum dolor sit amet, consectetur
    adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra,
    orci ligula finibus diam, sed tempus erat tortor id magna.
    Pellentesque habitant morbi tristique senectus et netus et malesuada
    fames ac turpis egestas.

Package initial Error: preset recursion limit exceeded (circular
reference?).

```

Legacy presets Named presets can reference other presets, enabling inheritance hierarchies; for compatibility and convenience, integer preset values from 1 to 6 invoke predefined behaviors for common use cases. Figure 2 represents the presets relative to the margin and text area.

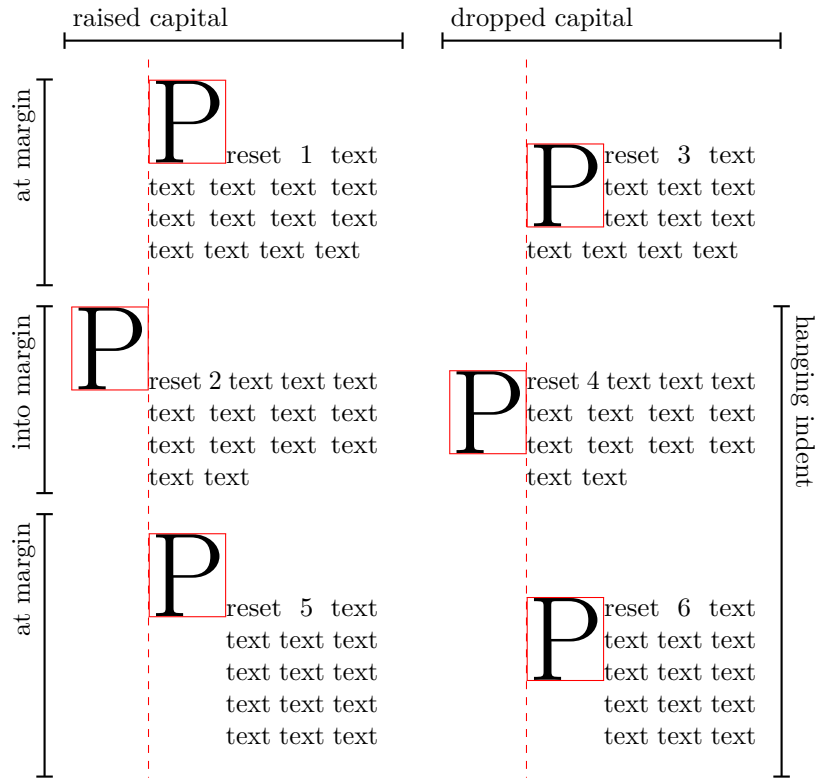


Figure 2: Presets arranged according to capital positioning relative to first line (either raised or dropped) and margin (at or into margin); red dashed line indicates the left margin, or, the left bound of the text space. Presets 1, 3, 5, and 6 are flush with the margin; presets 1, 2, and 5 are raised capitals; presets 3, 4, and 6 are dropped capitals; presets 2, 4, 5, and 6 result in hanging indentation relative to the capital, where the presets 2 and 4 are placed in the margin, and 5 and 6 are in the text area.

Preset 1 places the initial on the first baseline (a raised cap effect) with the entire initial width remaining within the text block. Text indents around it normally. Preset 2 similarly places the initial on the baseline but shifts it fully into the left margin, so text begins at the normal margin position, a marginal raised cap.

Preset 3 produces default drop cap behavior: the initial's top aligns with the first line's cap height, its bottom sits on line N 's baseline, and no special adjustments apply; this preset exists primarily for completeness, in that omitting the preset entirely produces the same result. Preset 4 combines standard drop cap positioning with marginal placement, shifting the initial into the left margin while maintaining its vertical position.

Presets 5 and 6 implement hanging indent patterns. Both configure the

after-region to maintain consistent indentation following the initial, creating a block of text that aligns at the initial's right edge. Preset 5 uses baseline positioning (raised cap style); Preset 6 uses standard drop cap positioning.

2.9 Global defaults

Every option parameter has a corresponding global default that can be set in the preamble. These defaults apply to all subsequent `\initial` calls unless overridden by explicit options. The counter defaults control integer parameters. Setting

```
\setcounter{initialDefaultLines}{4}
```

makes four-line initials the default throughout the document. Similarly, *initialDefaultDepth* and *initialDefaultAfterlines* establish defaults for their respective parameters.

Dimension defaults use `\setlength`. The offset defaults (`\initialDefaultVoffset`, `\initialDefaultHoffset`), indentation defaults (`\initialDefaultFindent`, `\initialDefaultNindent`), slope default, afterregion defaults, clearance defaults, and contour size all accept dimension specifications:

```
\setlength{\initialDefaultFindent}{0.2em}
\setlength{\initialDefaultNindent}{0.15em}
\setlength{\initialDefaultTclearance}{0.5\baselineskip}
```

Macro defaults use `\renewcommand*`. The scale factors (`\initialDefaultScale`, `\initialDefaultHstretch`, `\initialDefaultVstretch`) and colors (`\initialDefaultColor`, `\initialDefaultContourColor`) fall into this category:

```
\renewcommand*{\initialDefaultScale}{1.1}
\renewcommand*{\initialDefaultColor}{Blue}
```

2.10 Exposed dimensions

After an `\initial` call completes, three dimension registers contain the typeset initial's actual measurements: `\initialboxwidth`, `\initialboxheight`, and `\initialboxdepth`, and are the rendered size after all scaling and font calculations, decidedly not the input parameters.

```
\initial[lines=4]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitans morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

```
The initial width is \the\initialboxwidth.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus

erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The initial width is 36.122pt.

The exposed values are for conditional logic or subsequent positioning based on what actually rendered rather than what was requested. A document might well adjust following content based on whether the initial turned out other than expected.

Register	Description
<code>\initialboxwidth</code>	final width of the initial box
<code>\initialboxheight</code>	final height of the initial box
<code>\initialboxdepth</code>	final depth of the initial box

2.11 Debugging

Debugging remains relatively spartan. The `box` option renders a red frame around the initial, indicating its bounding box for alignment verification:

```
\initial[box]{L}orem ipsum dolor sit amet, consectetur adipiscing elit.  
Maecenas pellentesque, orci vitae blandit pharetra, orci ligula  
finibus diam, sed tempus erat tortor id magna. Pellentesque habitant  
morbi tristique senectus et netus et malesuada fames ac turpis  
egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The frame appears with zero padding, showing exactly what the package considers the initial's extent, to aid in diagnosing collision problems, font metric discrepancies, offset calibration issues; in this way, when the initial appears to collide with text, or floats in unexpected whitespace, the box shows whether the problem lies in the initial's positioning or the paragraph's shaping.

Option	Type	Default	Description
<code>box</code>	boolean	false	draw red frame around initial

2.12 Accessibility

Major revisions in 1.3.1. Ulrike Fischer noted that the accessibility features were faulty, in which content became displaced or unrendered. Here is a solution to that. Additio

When you typeset a decorative initial, the PDF contains either:

- a *scaled glyph*, in which the letter is present but may be transformed, styled, or positioned in ways that break text extraction, and so screen readers may skip it, read it out of order, or fail to associate it with the following text;
- an *image*, in which the ornamental letter carries no semantic content whatsoever, so to the PDF, it is indistinguishable from any other decorative graphic.

Without intervention, running `pdftotext` on a document with drop caps often yields results like “orem ipsum” instead of “Lorem ipsum”, whereby the `<L>` is entirely displaced relative to its paragraph.

Our proposed solution This package implements a two-layer approach to accessibility: invisible semantic anchor; decorative visual layer.

1. An invisible copy of the actual text (the letter, plus any **ante** content) is output in the margin just before the paragraph text; this anchor is rendered with zero opacity, and so is truly invisible on any background color, but creates real glyph content in the PDF stream. The anchor is wrapped with *ActualText*, which tells PDF readers “hey, this is the replacement text for accessibility purposes”.
2. The scaled glyph or ornamental image is output separately, marked with empty *ActualText*, so that text extraction tools ignore it. For Lua^AT_EX with `tagpdf`, it is additionally marked as an *artifact*, PDF terminology for decorative content that assistive technology should ignore. This layer provides the visual appearance, but contributes nothing to text extraction.

The result: we may extract and screen readers announce the correct letter, text extraction includes it in the proper position, and sighted users see only the decorative initial.

Usage For glyph initials, the process is automatic. The package uses the character itself as the actual text:

```
\initial{L}orem ipsum dolor sit amet, consectetur adipiscing elit.
Maecenas pellentesque, orci vitae blandit pharetra, orci ligula
finibus diam, sed tempus erat tortor id magna. Pellentesque habitant
morbi tristique senectus et netus et malesuada fames ac turpis
egestas.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

For image initials, you must provide `actualtext` explicitly, since the package cannot determine what letter an image represents:

```
\initial[actualtext=A]{deca.png}ve atque vale. Lorem ipsum dolor sit amet
, consectetur adipiscing elit. Maecenas pellentesque, orci vitae
blandit pharetra, orci ligula finibus diam, sed tempus erat tortor
id magna. Pellentesque habitant morbi tristique senectus et netus et
malesuada fames ac turpis egestas.
```



ve atque vale. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

A warning is issued if `actualtext` is omitted for image initials:

```
Package initial Warning: no actualtext for image initial.
Use actualtext=<letter> for accessibility.
Example: \initial[actualtext=W]{ornamental-W.pdf}
```

On ante text The `ante` option (for preceding punctuation like opening quotation marks) is automatically included in the accessibility output. When you write:

```
\initial[ante={``L}]{L}orem ipsum dolor sit amet, consectetur adipiscing
elit. Maecenas pellentesque, orci vitae blandit pharetra, orci
ligula finibus diam, sed tempus erat tortor id magna. Pellentesque
habitant morbi tristique senectus et netus et malesuada fames ac
turpis egestas.
```

“L orem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellentesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus erat tortor id magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

The `ActualText` is set to ``L (quotation mark plus letter), and text extraction correctly yields: "Lorem orem ipsum dolor sit amet.... This also works with image initials.

Verification To verify accessibility is working correctly:

1. run `pdftotext` and confirm initial letters appear in the output;
2. use `qpdf --qdf` to examine the content stream. You should see `ActualText` spans:

```
/Span<</ActualText(L)>>BDC
```

3. tools like PAC 2024 need to be checked; Adobe Acrobat's accessibility checker passes.

A Note on text extraction layout When using `pdftotext` with the `-layout` flag, the extracted text will reflect the visual indentation created by the drop cap:

```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Maecenas pellen-
    tesque, orci vitae blandit pharetra, orci ligula finibus diam, sed tempus
        erat tortor id magna. Pellentesque habitant morbi tristique senectus et
netus et malesuada fames ac turpis egestas.
```

This is expected behavior; the tool is accurately representing the document’s visual layout, where the first few lines are indeed indented to make room for the dropped initial.

Standard extraction and screen readers are unaffected; they read the content in logical order and produce clean output. The `ActualText` mechanism ensures the letter is correctly associated with the following text regardless of its visual position.

Some remarks on the technical details Our readability solution is a little hacky. The implementation uses the `accsupp` package to generate `ActualText` entries in the PDF. The invisible anchor is rendered with `\transparent{0}` from the `transparent` package, such that true invisibility is ensured regardless of background color. For Lua¹TeX users with `tagpdf` loaded, the decorative content is additionally wrapped as a PDF artifact using `\tag_mc_begin:n{artifact}`, providing proper structure tree integration for PDF/UA-2 compliance.

Tools like `pdftotext` use glyph coordinates, not PDF content stream order, to determine reading sequence. This creates a constraint: the semantic anchor must appear at coordinates that place it *before* the following paragraph text in left-to-right order. The implementation positions the anchor using `\llap`, which places it at a negative X coordinate (in the margin area). A tiny forward kern (0.1pt) is needed to ensure that the anchor’s effective position is immediately adjacent to the paragraph text. This satisfies the coordinate-based ordering while keeping the anchor visually outside the text area.

2.13 Troubleshooting

This following exposit on some issues encountered in use and testing. Other issues may exist, not accounted for here.

When the initial collides with preceding text, the automatic spacing calculation may be insufficient, or the preceding paragraph may not be terminating properly. Increasing `tclearance` adds explicit space; verifying a blank line or `\par` before `\initial` ensures proper paragraph separation. Unexpectedly high `voffset` values can raise the initial into collision range.

Collision with the following paragraph indicates the initial’s influence region exceeds the paragraph’s content. The `textlines` parameter addresses this by declaring the paragraph’s line count. Alternatively, explicit `bclearance` adds space regardless of line count. Initials with descenders may need `depth` increases.

Poor text flow around diagonal letters is for `slope` adjustment. Negative `findent` pulls the first line closer to the diagonal’s narrow top; positive slope shapes subsequent lines to follow the widening. The `box` option visualizes actual boundaries; the problem may be in the initial’s metrics rather than the shaping logic. Check the chosen font.

Image initials at wrong sizes imply image dimension or aspect ratio issues. The package scales to match computed height while preserving aspect ratio, so unexpectedly wide or narrow results indicate the source image’s proportions do not match expectations. Adjusting `lines` or `scale` compensates without modifying the source file.

Accented initials appearing too small are encountering the default diacritic-inclusive sizing. Enabling `basecap` sizes to the base letter, letting diacritics overflow; alternatively, increasing `lines` or using `scale` compensation accommodates full diacritics at larger size.

Presets failing to apply at all warrant checking the name spelling and ensuring `\newinitial` appears before the first invocation. Circular preset references trigger an error after 5 recursion levels; this is expected behavior.

Jagged contours may require larger `contoursize` values. Some fonts’ glyph geometries interact poorly with the contour algorithm, producing visible artifacts. Check the chosen font.

2.14 Some notes on implementation

The package builds paragraph shaping using $\text{\TeX}$ ’s `\parshape` primitive. Each `\initial` call constructs a specification listing the indentation and width for each line of the paragraph:

```
\parshape N indent1 width1 indent2 width2 ... indentN widthN
```

The line count N equals `lines` + `depth` + 1 for simple cases, or includes 100 additional lines when after-region formatting applies. The final specification repeats infinitely, governing all lines beyond those explicitly listed.

The initial itself renders via left overlap, placing it outside normal horizontal flow. The `\smash` command zeroes its vertical extent, preventing it from affecting line spacing. Together, these ensure the initial occupies its visual position without disturbing the paragraph’s geometric structure, so the paragraph behaves as though the initial were not there, except that `\parshape` indents text to make room for it.

Height computation normalizes across fonts using cap height measurement. The package typesets the target character (or `<X>` if `basecap` is enabled) at the computed height, measures the actual rendered result, and iteratively adjusts font size until rendered height matches target. The idea here is that iteration accounts for fonts with inconsistent or nonstandard vertical metrics, so the package adapts to what the font actually produces rather than trusting declared values.

The presets store option bundles using $\text{\TeX}$ ’s `\csname` to create named macros dynamically. Invoking a preset expands its stored options into the

current option-processing context. The recursion depth counter prevents infinite loops when presets reference each other, and erroring after 5 levels of expansion.

2.15 Comparison with Daniel Flipo’s work

Users familiar with `lettrine` or maintaining documents that use both packages will find suspiciously substantial overlap in vocabulary alongside meaningful differences in behavior.

The `lines`, `findent`, `nindent`, and `slope` parameters carry identical semantics between packages. The `ante` parameter exists in both, though `initial.sty` scales ante text by default while `lettrine` does not. The `ante*` variant for unscaled ante is unique to `initial.sty`. The `grid` option for baseline grid integration.

Scaling works differently. `lettrine`’s `loversize` adds to size (0.1 means 10% larger); `initial.sty`’s scale multiplies (1.1 means 10% larger). Vertical offset uses `voffset` in `initial.sty` versus `lraise` in `lettrine`, though functionality is the same. Horizontal adjustment differs more substantially: `lettrine`’s `lhang` specifies a proportion of the initial’s width to hang into the margin, while `initial.sty`’s `hoffset` specifies an absolute dimension. Both packages support image initials, though with different option syntax.

Several features exist only in `initial.sty`: the preset system with `\newinitiald` and inheritance; the style toggles (`bold`, `italic`, `smallcaps`, `slanted`); native color integration; the contour effect; the three-region paragraph model with after-region controls; the vertical clearance automation via `tclearance`, `bclearance`, and `textlines`.

`lettrine` offers features `initial.sty` lacks: the `\LettrineOptionsFor` mechanism for per-character default configurations; `novskip` for vertical skip adjustment. The default `nindent` differs significantly: `lettrine` uses 0.5em while `initial.sty` uses Opt. There is no RTL in `initial`. Documents migrating between packages may need adjustment.

Both can coexist in a document, as the `\lettrine` and `\initial` commands do not conflict.

3 Historical context

The decorated initial followed from structural necessity imposed by the codex format. The transition from scroll to bound book during Late Antiquity fundamentally altered reading mechanics: the scroll guided readers through continuous, linear engagement; the codex permitted random access and non-linear consultation, and so demanded visual signposts to orient the eye across discrete pages. In this context, the *litterae notabiliore*s ‘more noticeable letters’ arose as functional markers that projected into margins and broke justification lines to arrest the scanning reader [12]. The foundational typographic principle, that scale correlates with importance, originated here, and persists even in modern HTML heading hierarchies.

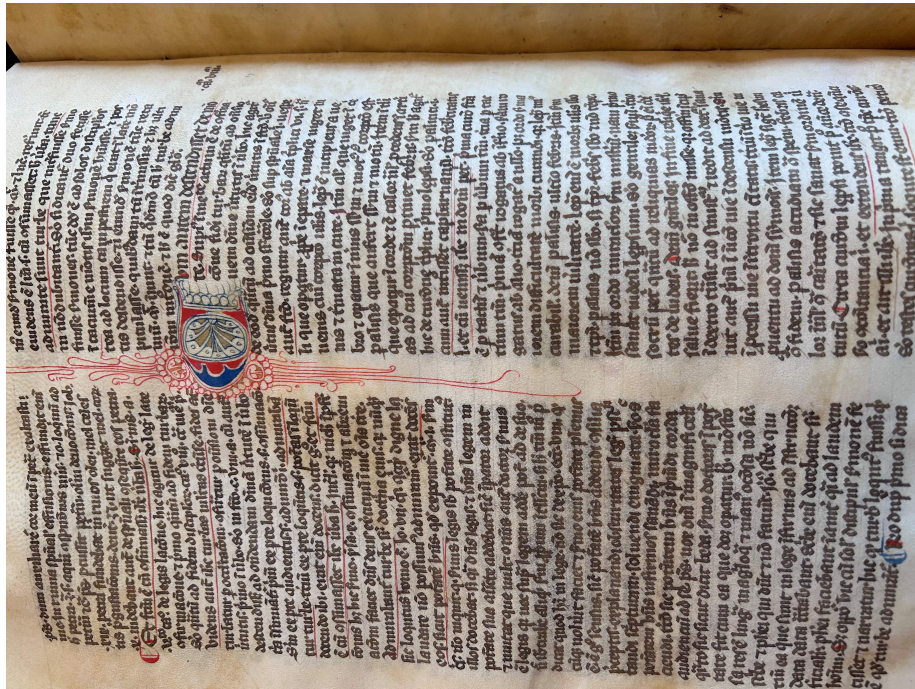


Figure 3: Duns Scotus, John, approximately 1266–1308. *Postilla super Mathaeum*. 15th century. 091.D926p. Image source: USC Libraries.

Color reinforced structural grammar; indeed, the term rubric derives from *rubrica* ‘red ochre’, denoting the red ink, frequently *minium* ‘red lead’, used to highlight initials and section headings against carbon-black body text [35]. This was what paleographer M.B. Parkes termed a “grammar of legibility”: graphic conventions structuring textual transmission [29]. The rubricated initial allowed location of section beginnings without parsing preceding text.

Script hierarchies often accompanied these enlarged letters. A manuscript might present titles in Rustic Capitals, incipits in Uncials, and body text in minuscule, creating a “diminuendo” effect, a gradual reduction from monumental that funneled attention from the initial to the text block [1, 7]; this required substantial planning: scribes calculated spacing to accommodate irregular expanding forms while maintaining visual rhythm.

3.1 Insular tradition

In the monasteries of Ireland and Northumbria (c. 600–900 CE), the initial became a liturgical object. The Book of Kells (c. 800) and Lindisfarne Gospels exemplify this transformation, where “interlace initials” achieved complexity that subordinated legibility to symbolic density [28]. The Chi-Rho page of Kells (Folio 34r), for example, expands the Greek monogram of Christ into a full-page explosion, a “carpet page”, signaling presence and requiring contemplative pause before engagement.



Figure 4: Chi Rho page (folio 34R) of the Book of Kells (c. 800). The letters Chi, Rho and I are an abbreviation of “Christ”. Image source: Wikimedia Commons.

Following colossal initials consuming the page, subsequent letters downward in size and complexity, rubricated, outlined, or color-filled, smoothed the transition

between sacred invocation and narrative flow. Furthermore, these manuscripts pioneered word separation and punctuation, enhancing legibility for non-native speakers [16, 34].

3.2 Gothic systematization

The migration of book production from monastic scriptorium to urban workshop (*stationarius*) during the 12th and 13th centuries demanded standardized visual language. In university centers of Paris, Bologna, Oxford, the system known as *ordinatio* developed, providing a visual index for scholastic texts increasingly viewed as compilations of discrete parts requiring non-linear access [36, 10]. The hierarchy was functional navigation, where material cost directly correlated with textual authority [10, 30].

Professional workshops calculated manuscript costs by initial type and quantity. A historiated initial required specialist painters and expensive materials, such as gold leaf, lapis lazuli, while flourished initials could be executed by scribes or specialized *fleuristes*, using standard colored inks. Lombard capitals represented the cheapest decorative unit [38]. Patron budgets and textual structure aligned: significant divisions received expensive decoration.

Table 2: The Gothic hierarchy of initials

Level	Technical term	Material composition	Textual function
Major	historiated; inhabited	gold leaf; gesso; multiple pigments	incipit of books, major sections
Middle	<i>champie</i> , foliate, <i>florissa</i>	gold leaf; pigment (rose, blue); white lead	chapters; prologues
Minor	<i>florissa</i> , <i>fleuronné</i>	penwork; colored inks (red/blue)	Psalms; lessons; stanzas
Minimal	Lombardic, versal	solid pigment (red, blue); no gold	verses; paragraphs

This hierarchy functioned as cognitive aid. Visual weight provided instant cues to textual landscape: major initials signaled fundamental resets; minimal provided beats of verses or legal points [27, 26].

Major initial; historiated and inhabited The apex of decoration, the **historiated initial**, contained narrative scenes or identifiable figures within the letter’s counter. The historiated initial was part of the text string, essentially a letter that had become a picture that served as visual exegesis: the initial of Psalm 1 (*Beatus vir*) contained King David playing the harp, establishing authorship and liturgical tone visually.

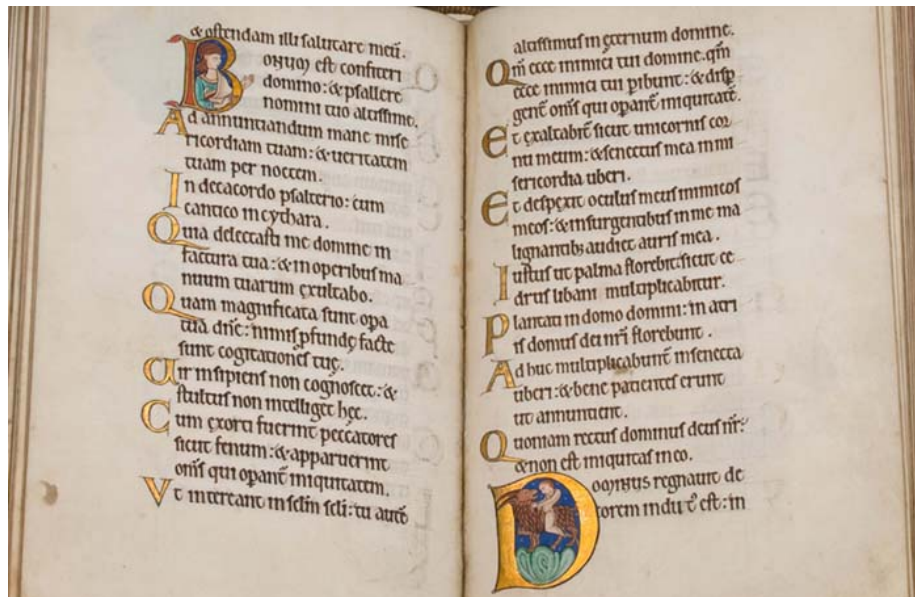


Figure 5: Hunterian Psalter (c. 1170). Folios 120v–121r. Historiated initial (psalm 91) and historiated initial <D> (psalm 92). Image source: University of Glasgow Special Collections.

The specificity was crucial; a depiction of St. Margaret emerging from the dragon functioned as a visual label identifying a prayer text for skimming readers [35].



Figure 6: Aberdeen Bestiary (c. 1100–1200). St. Margaret and the dragon, forming the initial. Image source: trappedinthescriptorium.

A necessary distinction separates historiated from inhabited initials: inhabited initials contained living figures as human, animal, or hybrid that did not enact specific narratives; a letter <Q> containing a dragon biting its tail or a figure climbing a vine, absent specific character identification, was *inhabited* rather than historiated. In strict hierarchies, historiated outranked inhabited, though both occupied the “Major” tier relative to abstract or foliate decoration.

Production involved complex workflow. Scribes left blank indented spaces six, eight, ten lines deep, with tiny guide letters in margins or voids instructing illuminators which letter to paint [12].

Middle initial: champie and foliate The *champie initial* became the standard marker for chapter divisions in Paris and Low Countries workshops. The term derives from French *champ* ‘field’: a gold letter (burnished gold leaf) on colored paint fields (alternating rose and blue), or vice versa, decorated with delicate white filigree patterns (*rinceaux*) [14]. The *champie* created clear visual distinction from the Major tier (lacking figures) and Minor tier (employing gold), positioning gold as the middle-tier indicator, more precious than ink, less communicative than picture.



Figure 7: Le Mans, Médiathèque Louis Aragon, MS 354. Image source: Lancelot-Grail Project by DIAMM. University of Pittsburgh.

In the earlier Romanesque period (11th–12th centuries), distinction between Major and Middle is ambiguous. Decorated initials featured gymnastic figures (men or beasts forming letter bodies) or zoomorphic forms. As Gothic style matured, this evolved into the *foliate initial* with stylized ivy leaves or vines.

Minor initial: pen-flourished The *littera florissa* or *lettre fleuris* marked the shift to pen and ink. Typically bichromal, flourished initials employed standard red and blue stationer’s inks: blue letters surrounded by red penwork, red by

blue. The flourish consisted of abstract linear patterns as spirals, bead-lines, hatch-marks, framing letters and extending into margins, serving the practical function of catching scanning eyes and anchoring text blocks [3, 5].

The puzzle initial, common in university Bibles and legal texts from Paris and Oxford, composed interlocking red and blue pieces separated by thin white parchment lines [19]. These occupied the upper Minor tier, sometimes serving as Middle tier initials, where gold was prohibitively expensive.



Figure 8: Manuscript cutting, initial <U>, Germany, Rhineland, c. 1450. Image source: Fragmented Illuminations: Medieval and Renaissance Manuscript Cuttings at the V&A.

Flourishing style tends to indicate manuscript origin [19]:

Region	Characteristics
Paris	delicate, thread-like flourishes
Bologna	“heads of barley” (rows of small circles); long bead-like extensions
England	bolder, heavier flourishes with “saw-tooth” patterns and “J-borders”; column length

Minimal initial: Lombard capitals The term “Lombard” in this context refers to script style, not primarily to Lombardy. Lombardic capitals derive from Uncial forms: rounded; bulbous; composed of curved sweeping strokes contrasting sharply with angular, fractured Gothic blackletter [39]; they were almost universally mono-colored (alternating red and blue), solid letters lacking internal puzzle division or external filigree.



Figure 9: Lombardic capitals. Ambraser Heldenbuch, fol. 75v, c. 1516). Image source: Wikipedia.

3.3 Regional and chronological divergences

Romanesque In the earlier Romanesque period, hierarchy was less rigid. Gymnastic initials of men and beasts forming letters themselves blurred Major and Middle distinctions. A decorated initial in a Romanesque Bible might match historiated initials in size and complexity, relying on intricate interlace rather than narrative to convey status [28].

Italian humanist Humanist scholars in 15th-century Italy, seeking classical revival, rejected Gothic hierarchy; indeed, Humanists applied pejorative language to the Gothic system broadly [9].

Writing was replaced with epigraphic capitals: square, seriffed letters modeled on Roman inscriptions like Tajan’s column [40], which became especially celebrated in later typography scholarship.

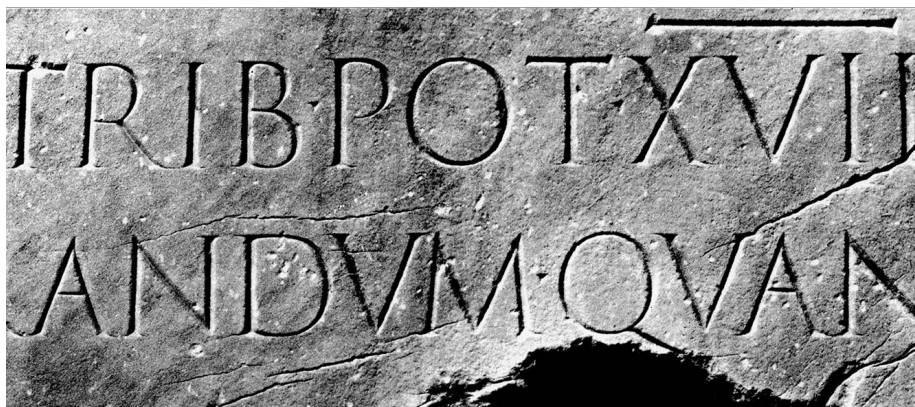


Figure 10: Excerpt from Tajan’s column.

Instead of Gothic *champie* or ivy-leaf decoration, Humanist manuscripts employed *bianchi girari* (white vine) decoration, using white parchment to form vine stems against colored backgrounds, imitating 11th-century Tuscan manuscripts mistakenly believed classical [20]. Humanist manuscripts often reverted to single-column layout, rejecting dense Gothic two-column university format.



Figure 11: Life of Alphonso VI, King of Aragon and Naples (1416–1458). Image source: Wikipedia.

Bologna school Legal manuscripts from Bologna developed a distinct hierarchy. Text of law (center) and gloss (surrounding) required separate initial hierarchies to prevent confusion. Lombard capitals often appeared in glosses while main text utilized flourished or puzzle initials, maintaining visual separation between Authority (Law) and Commentary (Gloss) [19, 22].

3.4 The Incunabula

Gutenberg’s movable type (1450s) did not immediately render hand-painted initials obsolete. Early printed books (*incunabula*) occupied liminal space between unique manuscript and mass commodity. Printers seeking to compete with prestigious manuscript markets produced books mimicking handwritten texts: body text printed in black ink with blank chapter spaces left for rubricators to add initials by hand [43].

Printers often printed small guide letters (directors) in blank spaces, intended to be painted over and disappear beneath opaque pigments. In many surviving incunabula, rubrication was never completed, due to cost, production speed, or

owner indifference, leaving blank spaces or guide letters visible in large white squares, in a sense, archaeological evidence of transition-period economics.

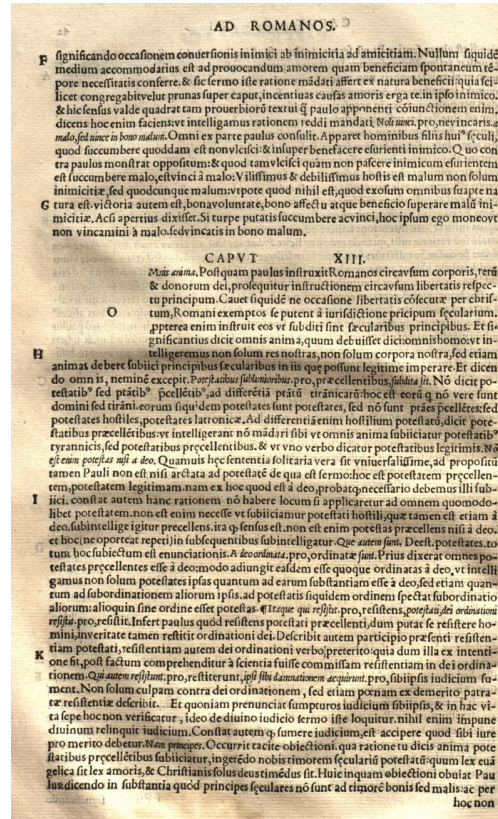


Figure 12: Epistolae Pauli et aliorum Apostolorum ad graecam veritatem castigate. Image source: [21], from Thomas Fisher Rare Book Library, University of Toronto.

Hand-finishing bottlenecked mass production. Printed editions of hundreds took weeks; individual rubrication took months. This economic friction drove technological development of printed initials [21, 4].

Erhard Ratdolt and woodcutting Erhard Ratdolt, a German printer working in Venice (1476–1486) and later Augsburg, transformed decorated initial production: he developed woodcut initial sets, which are relief blocks carved from wood, lockable into chases alongside metal type [43, 37]. His *Calendarium* (1476) and *Euclid's Elements* (1482) showcased white vine (*bianchi girari*) initials, imitating Italian Humanist manuscript style.

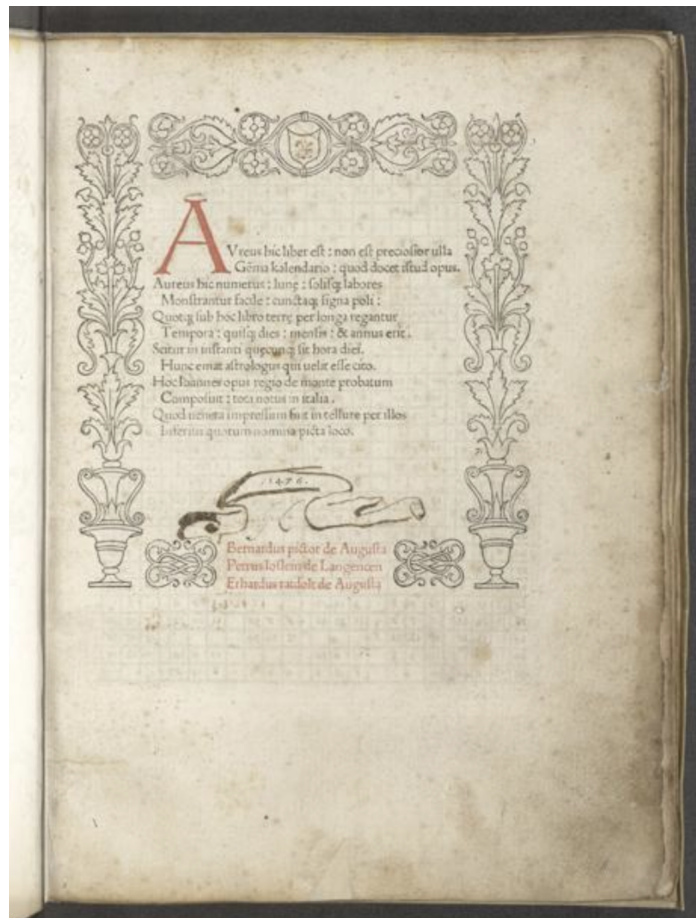


Figure 13: Erhard Ratdolt, *Calendarium* (1476). Image source: Library of Congress.

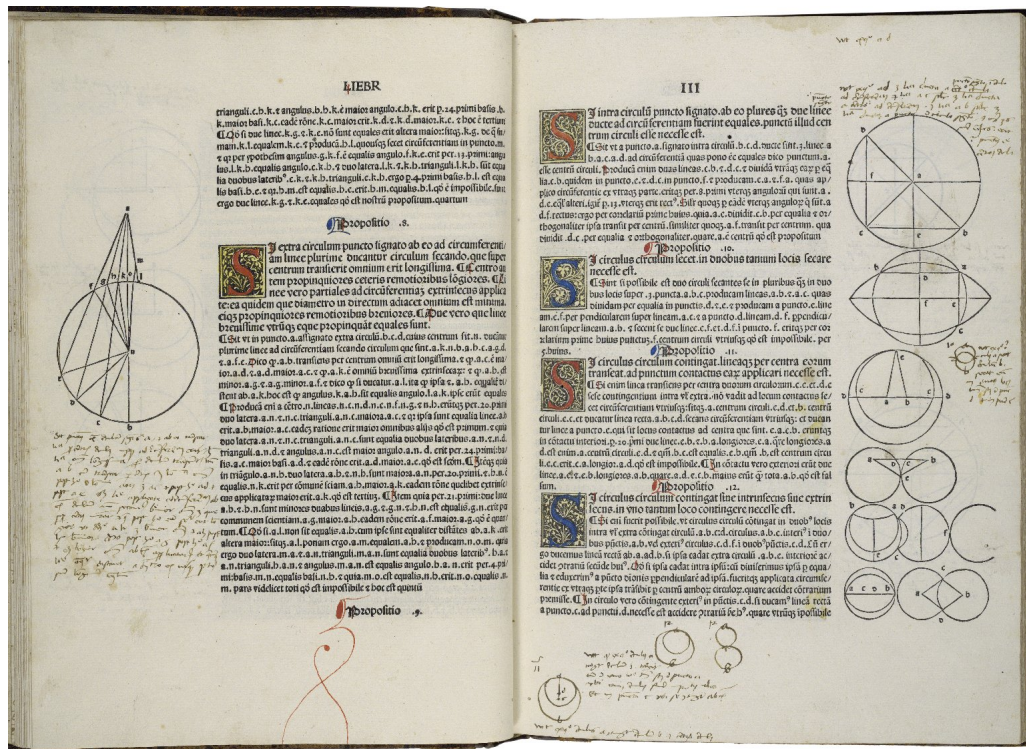


Figure 14: Erhard Ratdolt, *Euclid's Elements* (1482). Image source: Wikipedia.

Unlike heavy solid Lombardic forms, Ratdolt's initials were delicate outlines suitable for hand-coloring or, significantly, left as black-and-white line art. The illuminated initial became mass-producible; Ratdolt also experimented with multi-color printing, attempting red-ink initials during separate press passes, though registration difficulties meant most printers preferred black woodcut initials with optional manual coloring.

3.5 Renaissance: Geoffroy Tory and the letter as philosophy

By the early 16th century, the printed book had matured, and initial design became subject of inquiry. Geoffroy Tory, in *Champ Fleury* (1529), elevated letter-form design to philosophical pursuit [6]. Rooted in Renaissance Neoclassicism, Tory argued Roman capital proportions were divinely inspired and derived from the human body, superimposing letters over grids and human figures [44].

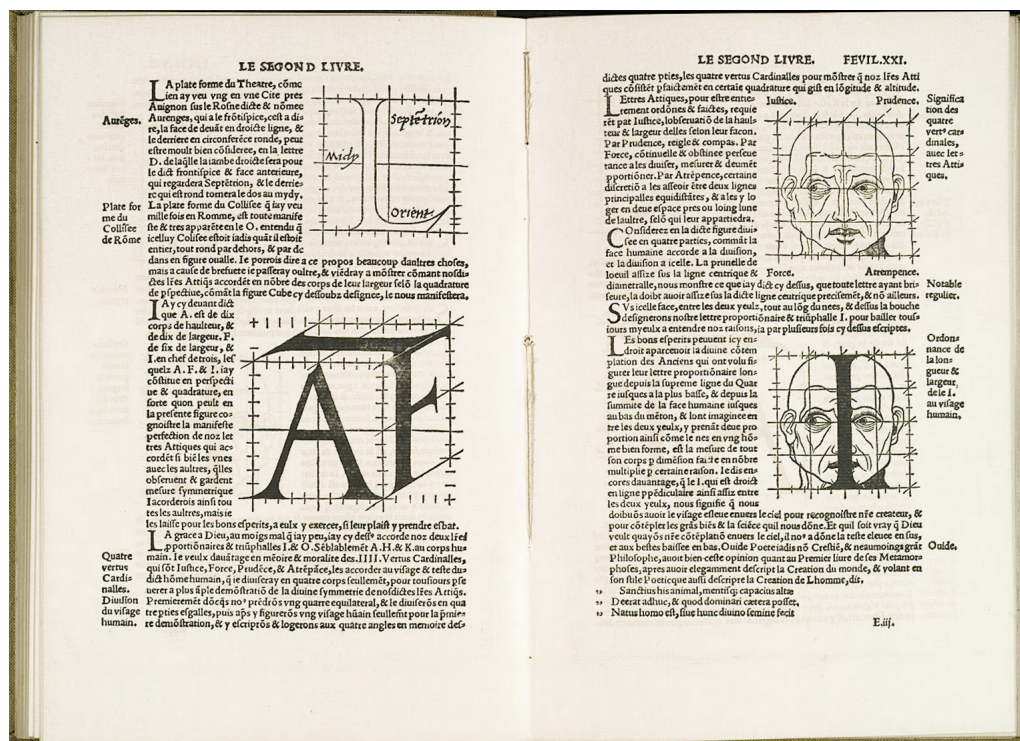


Figure 15: Geofroy Tory, *Champ fleury*, au quel est contenu l'art et science de la deue et vraye proportion des lettres attiques, quon dit autrement lettres antiques et vulgairement lettres romaines, proportionnees selon le corps et visage humain, Paris, 1529. Bibliothèque nationale de France. Image source: [44].

Tory's *criblé* initials (letters against white-dot backgrounds resembling sieves) introduced lighter aesthetics to French printing. His work standardized Roman capitals in French typography.



Figure 16: Geofroy Tory, *criblé* initial <G>.

3.6 The Factotum

As printing expanded, the factotum initial emerged. From Latin *fac totum* ‘do everything’, the factotum was a decorative woodblock frame with a central hole (*mortise*) into which any standard metal type letter could be inserted [25]. A single decorative border served every alphabet letter.

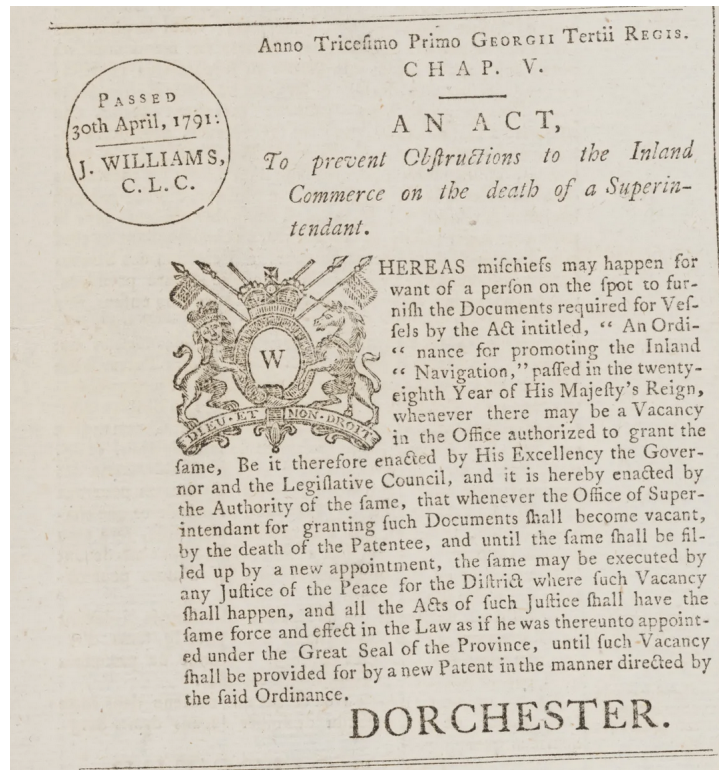


Figure 17: Factotum initial <W>. *La Gazette de Québec*, 1791. Any letter could be used with the same surrounding decoration block. Image source: [31].

Economically brilliant but aesthetically compromised, the factotum’s inserted type rarely matched border style or weight, spacing often awkward [41]. Factotums became hallmarks of rapid production and initial commoditization as reusable mechanical part [31].

3.7 Neoclassical decline; Arts & Crafts revival

The late 18th century witnessed radical typographic shift under Bodoni and the Didot family. Their “Modern” (Didone) typefaces, characterized by extreme thick-thin contrast, hair-thin unbracketed serifs, severe vertical axis, proved hostile to traditional decorated initials [17, 42]. Heavy woodcut initials looked clumsy against razor-sharp Bodoni hairlines. Neoclassical ideals favored purity over Rococo ornamentation or Gothic density [8]. The decorated initial fell from favor, replaced by plain drop caps in matching fonts; the white page luxury.



Figure 18: William Morris, *News from Nowhere: Or, an Epoch of Rest* (London: Kelmscott Press, 1892); Pequot Library Special Collections. Image source: Wikipedia.

William Morris's Arts and Crafts movement (1890s) directly reacted against this industrial grayness. Morris founded Kelmscott Press (1891) to produce books as art objects, returning to woodcuts visually heavy enough to balance his dense "Troy" and "Golden" typefaces [24, 33]. Morris treated double-page spreads as design units, initials as complex border-and-frame systems [33]. The Kelmscott style reinvigorated initial art, inspiring private presses and eventually filtering into commercial graphic design.

3.8 Digital typography

Desktop publishing democratized drop caps, but lost traditional nuance. Standard software uses rectangular bounding boxes, ignoring irregular letter shapes (<A>; <V>; <L>); professional typography still requires contour wrapping around actual letter shapes, demanding manual path-drawing in software like Adobe InDesign (more on this below).

HTML and CSS treat text as fluid medium reflowing by screen size, complicating fixed initial-text relationships. The `::first-letter` CSS pseudo-element

allowed distinct first-character styling, but the newer initial-letter property provides magazine-quality control over sink (lines dropped) and raise (height above baseline), though browser support remains fragmented (see <https://developer.mozilla.org/en-US/docs/Web/CSS/Reference/Properties/initial-letter>).

Digraphs present an additional complication. The Dutch <IJ> digraph illustrates digital internationalization complexity. In Dutch, <IJ> functions as a single letter for the phoneme /ɪ/; dropping only <I> from <IJsselmeer> while leaving <J> normal-sized is typographically incorrect. Historic metal type included ligatured <IJ> sorts; modern browser specifications are beginning to address such linguistic features, allowing selection of both characters as initial units.

4 Initials in modern typography

LaTeX’s `lettrine` package remains the most parametrically sophisticated solution for beautiful typesetting of initials; Adobe InDesign provides intuitive visual interface, but lacks programmatic flexibility; CSS’s *initial-letter* property, despite achieving 91% browser coverage in 2025, still lacks Firefox support, and offers no contour-fitting capability. Additional technical challenges across all systems stem from inconsistent font metric standards established decades ago, which, especially in the case of LaTeX, must contend with being digital objects at all, as opposed to printed objects.

4.1 Initials in LaTeX: `lettrine.sty`

DANIEL FLIPO authored the `lettrine.sty` package, which has evolved through over 25 years of development to the current v2.83 (released December 6, 2025) [15]. The package represents the culmination of earlier efforts, including `dropcaps.sty` [23], the `drop.sty` package [11], and `dropping.sty` [13], all of which CTAN now marks as superseded by `lettrine`.

The package’s parameter system provides granular control. Version 2.60 (January 2024) introduced right-to-left typesetting compatibility with LuaLaTeX and XeLaTeX. Version 2.81 added the `actualtext` parameter for PDF tagging, addressing accessibility requirements for tagged PDF output. For further reading on the package’s design philosophy, the documentation cites [18, 32] as authoritative French typographic references.

4.2 Initials in LaTeX: `novel.cls`

Robert Allgeyer’s `novel` document class, designed specifically for print-on-demand fiction publishing, provides an alternative `\dropcap` command distinct from `lettrine` [2]. The class prioritizes PDF/X compliance and baseline grid critical requirements for commercial printing that general-purpose LaTeX classes often handle imperfectly.

`novel.cls` enforces strict baseline alignment throughout the document, and its drop cap implementation respects this constraint natively; the `lettrine` package added grid support later via the `grid` option, but it remains optional rather than foundational.

`novel.cls` works exclusively with Lua^AT_EX and requires TeXLive 2020 or later; Allgeyer marked it as deprecated but functional through TeXLive 2024. The documentation notes that “works of popular fiction rarely use drop caps”, and recommends special first characters instead, as they “present fewer layout problems, and are less distracting”.

4.3 InDesign initial letters

Adobe InDesign offers visual control, but limited automation; throughout, see <https://helpx.adobe.com/indesign/using/drop-caps-nested-styles.html>. InDesign provides drop cap controls in three interface locations: the *Paragraph* panel; the *Control panel*’s paragraph formatting section; the *Drop Caps and Nested Styles* dialog accessible through paragraph style options. The two parameters, *Drop Cap Number of Lines* and *Drop Cap One or More Characters*, control vertical extent and horizontal scope, respectively.

The *Align Left Edge* option, added in InDesign CS3, addresses a persistent typographic annoyance: characters appear indented from the margin due to their side bearings (the invisible space flanking each glyph); enabling this option uses the actual left edge of the letterform, rather than its full advance width. The *Scale for Descenders* option reduces the size of descending drop caps (<Q>) to prevent overlap with text below, though this noticeably shrinks the character. Nested styles extend drop cap functionality by embedding character styles within paragraph styles, automating formatting sequences after the initial.

InDesign’s limitations, compared to programmatic design, include: inability to dynamically calculate drop cap character counts for varying word lengths; no native way to apply different drop cap styles based on content analysis; no contour-fitting for diagonal letters without converting to anchored graphics and manually editing wrap boundaries.

4.4 Other typesetting

QuarkXPress has offered drop caps since at least version 5.0, with parameters including character count (1–127), line count (2–16), and percentage-based sizing. The 2023 version added syllable-based drop cap support for Indian languages including Hindi, Bengali, Tamil, and Telugu, addressing scripts where character boundaries differ from Latin conventions. See <http://quarkxpress.helpmax.net/en/text-and-typography/creating-drop-caps/> and https://www.quark.com/documentation/quarkxpress/2023/english/User%20Guide/_299d82f7-f44b-44d1-b6d9-3bfe69b70f.html.

CSS’s *initial-letter* property (CSS Inline Layout Module Level 3, W3C Working Draft) provides the most promising web-native solution. The syntax *initial-letter : 3 2* creates a three-line-tall initial with its baseline at

line 2. Safari first supported the property with the *-webkit-prefix* in version 9 (September 2015), Chrome and Edge added partial support in version 110 (February 2023), but Firefox remains unsupported <https://caniuse.com/css-initial-letter>. Before *initial-letter*, developers relied on floating the *:first-letter pseudo-element*, a technique with significant cross-browser inconsistencies. See <https://developer.mozilla.org/en-US/docs/Web/CSS/Reference/Properties/initial-letter>.

Adobe’s Web Platform team created *dropcap.js* to provide proper baseline alignment through JavaScript, calling *window.Dropcap.layout(elements, 3)* to create three-line drop caps. The library requires explicit span markup, because the CSS Object Model provides no write access to *:first-letter* pseudo-elements. The documentation notes: “assumes Western content”, and requires fonts to be fully loaded before layout calculation; see <https://github.com/adobe-webplatform/dropcap.js/>.

Microsoft Word offers drop caps through *Insert* → *Text* → *Drop Cap*, with two position options (*Dropped* or *In Margin*), font selection, lines to drop (1–10), and distance from text. The feature uses frames internally, and cannot be applied within tables or text boxes. See <https://office-watch.com/2021/drop-caps-in-word/>.

Apple Pages added drop cap support in version 10.0 (2020), providing six preset styles, background shape options with customizable corner radius, and an outdent control for positioning relative to the margin. The implementation works across macOS, iOS/iPadOS, and Pages for iCloud. See <https://www.cultofmac.com/how-to/how-to-use-drop-caps-pages>.

Jessica Hische’s *Daily Drop Cap* project (see <http://www.dailydropcap.com/about>) has produced complete illustrated alphabets, and led to the Penguin Drop Caps series, a contemporary revival of decorated initials in commercial publishing (See <http://www.dailydropcap.com/about>).

Fundamental technical challenges remain unsolved in any universal sense. Baseline alignment represents the fundamental challenge: drop caps must position their tops at the first line’s cap height while their bottoms rest on a specific lower baseline. Font metrics calculation complicates sizing across platforms; the SIL Font Development Best Practices documentation notes that “winAscent should accommodate tallest accented character”, a requirement many fonts fail to meet, causing clipping of diacritical marks; indeed, diacritics may certainly extend above standard cap height, disrupting alignment calculations at all (see, for a full discussion of the problem in HTML: <https://www.codestudy.net/blog/how-to-correctly-align-a-drop-cap-in-html-css/>).

Paragraph shaping around irregular letter shapes (<A>; <V>; <W>; <T>; <L>; <Y>) demands contour fitting. *lettrine*’s slope parameter enables per-line indent adjustment, InDesign requires converting drop caps to anchored graphics with manually edited wrap boundaries, and CSS offers no real contour-fitting capability (at this time: January 15, 2026).

The choice between glyph-based and image-based initials involves trade-offs: glyph-based initials scale, remain searchable, and support screen readers; image-based initials enable complex illumination, multi-color designs, and photographic

elements impossible within font technology, but must be tagged for readability.

Each system implements workarounds suited to its environment: L^AT_EX’s per-character configuration files; InDesign’s manual text wrap editing; CSS’s fallback stacking. Workarounds trace directly to decisions made when digital font formats were standardized in the 1980s and 1990s, before drop caps received serious consideration. Contemporary writers, therefore, must work within those inherited constraints; the hope is, that we might achieve results that would satisfy a medieval rubricator, and acknowledge that true illumination is an artistry that otherwise remains beyond pure automation.

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A Parameter reference

The following inventory lists all parameters accepted by the `\initial` command, organized by function. Each entry notes the parameter type, default value, and corresponding global override where applicable. Note: due to the information overload here, we refrain from the otherwise more convenient table format, and elect a list.

A.1 Dimensional parameters

lines

integer default: 3 global: `\initialDefaultLines` (counter)

Number of text lines the initial spans vertically. Initial’s top aligns with first line’s cap height; its bottom rests on line *N*’s baseline.

depth

integer default: 0 global: `\initialDefaultDepth` (counter)

Additional lines of indentation below initial’s visual extent; accommodates descenders or decorative flourishes.

offset

length default: 0pt global: `\initialDefaultOffset`

Horizontal shift of initial; positive values move rightward into text block; negative values move leftward into margin.

voffset
length default: 0pt global: \initialDefaultVoffset
 Vertical shift of initial; positive values raise it; negative values lower it.

findent
length default: 0pt global: \initialDefaultFindent
 Gap between initial’s right edge and first line of text.

nindent
length default: 0pt global: \initialDefaultNindent
 Gap between initial’s right edge and subsequent lines (lines 2 through lines + depth).

gap
length default: 0pt global: —
 Additive adjustment applied to both **findent** and **nindent**; provides shorthand for uniform spacing changes.

slope
length default: 0pt global: \initialDefaultSlope
 Per-line indent increment (for diagonal letters). Positive values increase indentation on each successive line (for letters widening downward like <A>); negative values decrease it (for letters narrowing downward like <V>).

scale
float default: 1 global: \initialDefaultScale (macro)
 Overall size multiplier applied to computed initial height. A value of 1.2 produces an initial 20% larger than lines alone would yield.

hstretch
float default: 1 global: \initialDefaultHstretch (macro)
 Horizontal stretch factor. Values greater than 1 widen the initial; values less than 1 compress it; combines multiplicatively with **scale**.

vstretch
float default: 1 global: \initialDefaultVstretch (macro)
 Vertical stretch factor. Values greater than 1 heighten the initial; values less than 1 compress it; combines multiplicatively with **scale**.

A.2 Three-region paragraph model

afterlines
integer default: 0 global: \initialDefaultAfterlines (counter)

Flag enabling post-initial region control. When greater than zero, the package generates 100 additional shaped lines using **afterindent** and **afterslope**.

afterindent

length default: 0pt global: \initialDefaultAfterindent

Base indentation for lines following initial region; active only when **afterlines** > 0.

afterslope

length default: 0pt global: \initialDefaultAfterslope

Per-line indent increment for post-initial region; enables gradual return toward margin or other transitional effects.

A.3 Typography

font

command default: (inherit) global: —

Font-switching command for initial. Accepts any valid font selection, e.g., **\sffamily**, **\fontspec{...}**.

bold

boolean default: false global: —

Apply **\bfseries** to initial.

italic

boolean default: false global: —

Apply **\itshape** to initial.

smallcaps

boolean default: false global: —

Apply **\scshape** to initial.

slanted

boolean default: false global: —

Apply **\slshape** to initial.

color

color name default: (inherit) global: \initialDefaultColor (macro)

Color of the initial; accepts any color name recognized by **xcolor**; empty value inherits document color.

contour

boolean default: false global: —

Enable contour/outline effect around initial character.

contourcolor
color name default: `white` global: `\initialDefaultContourColor`
 (macro)

Stroke color for contour effect.

contoursize
length default: `0.3pt` global: `\initialDefaultContourSize`

Stroke width for contour effect.

A.4 Ante and post text

ante
string default: (empty) global: —

Text placed before initial, scaled to approximately 70% of initial height, raised with initial, hanging in margin. A fixed 0.2em gap separates ante from initial.

ante*
string default: (empty) global: —

Text placed before initial at body text size, positioned at baseline, hanging in margin; no scaling applied.

post
style spec default: `textsc` global: `\initialDefaultPost` (macro)
 Styling applied to optional trailing argument. Accepts comma-separated values (multiple in curly brackets): `textsc`; `textbf`; `textit`; `textsl`; `texttt`; `none`; `normal`.

A.5 Vertical clearance

tclearance
length default: `0pt` global: `\initialDefaultTclearance`

Fixed space added above initial paragraph, supplementing automatic collision prevention.

bclearance
length default: `0pt` global: `\initialDefaultBclearance`

Fixed space added below initial paragraph; combines additively with calculated spacing from `textlines`.

textlines
integer default: `0` global: —

Declares paragraph's line count for automatic bottom spacing calculation. When specified, the package computes

$$\text{space} = \max(0, (\text{lines} + \text{depth} - \text{textlines})) \times \text{baselineskip}$$

and adds this below the paragraph.

grid

boolean default: `false` global: `\initialDefaultGrid` (conditional)

Round vertical space above the paragraph to an integer multiple of `\baselineskip` for baseline grid alignment; set globally via `\initialDefaultGridtrue` or `\initialDefaultGridfalse`.

A.6 Diacritics and accessibility

basecap

boolean default: `false` global: `\initialDefaultBasecap` (conditional)

Size initial to base cap height (calibrated against `<X>`), allowing diacritics to overflow above initial region; set globally via `\initialDefaultBasecaptrue` or `\initialDefaultBasecapfalse`.

actualtext

string default: (auto) global: —

Replacement text for PDF accessibility tagging. For text initials, defaults to character itself. Required for image initials, which carry no inherent semantic content.

A.7 Presets and debugging

preset

name or integer default: — global: —

Invoke a named preset defined via `\newinitial`, or a legacy numeric preset (1–6); user-specified options override preset values.

box

boolean default: `false` global: —

Render a red frame around initial’s bounding box for debugging alignment and collision issues.

A.8 Exposed dimensions (read-only)

The following dimension registers are populated after each `\initial` call, and reflect actual rendered measurements:

`\initialboxwidth`

Final width of initial box after all scaling.

`\initialboxheight`

Final height of initial box after all scaling.

`\initialboxdepth`

Final depth of initial box after all scaling.

A.9 Commands

`\initialoptions`*character or image*`post text`****

Primary command; creates an initial at paragraph start. The first optional argument accepts key-value pairs; the mandatory argument specifies the letter or image file; the trailing optional argument provides styled run-in text. Controls paragraph behavior if set accordingly.

`\newinitial`*name*`options`****

Define a named preset for reuse. The preset's options apply when invoked via `preset=name`, and can be overridden per-call.